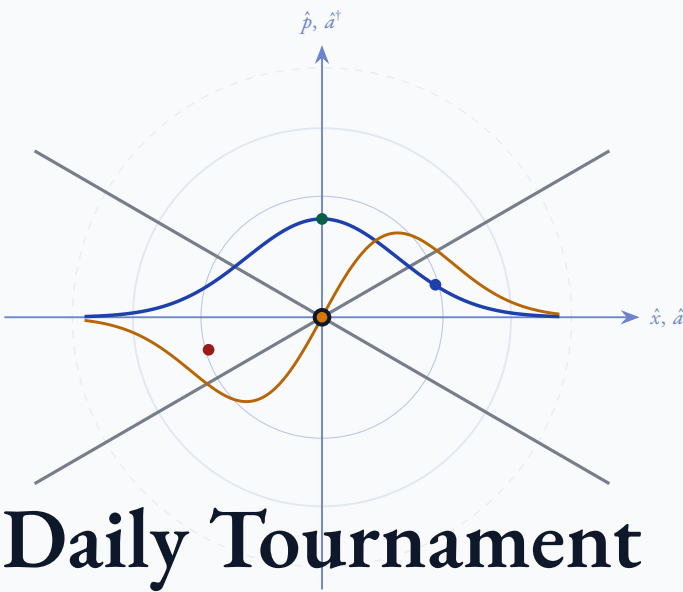




Daily Tournament

GRAND ARENA OF THEORETICAL PHYSICS

“Today’s contestants are ready: 604 Variable-Upper Integral $\times$ 811 Harmonic Oscillator”

Closed-book · Pen-and-paper derivation · No shortcuts

Date	2026-09-01 Week 1 (Day 2)
Modules	604: Variable-upper limits & Taylor 811: Harmonic oscillator wave mechanics
Error Protocol	5-Factor: Knowledge Modelling Logic Calculation Time
Target	Standard 5.5–6.5 h Minimum Defence 3.5 h

Theoretical Physics Training · Daily Tournament

2026-09-01 | Week 1 (Day 2) | Phase: Foundation diagnostic

§1 Tactical Briefing & Warm-Up Retrieval

- 604 今日靶向: 变上限积分导数与泰勒展开综合应用。熟练运用 Leibniz 变限求导定理及被积函数高阶展开定阶, 攻克 M1 积分分子与等价替换分母之极限。
- 811 今日靶向: 一维简谐振子降算符微分表象 $\hat{a}\psi_0 = 0$ 解一阶微分方程求高斯基态波函数; 利用阶梯代数求解偶极跃迁矩阵元 $\langle m|\hat{x}|n\rangle$ 与测不准关系 $\Delta x \Delta p = \hbar/2$ 。
- Supporting: 英语精读 2015 Text 1, 掌握长难句三刀切脱水与同义改写; 政治学习马原辩证法对立统一规律。

Momentum Defence Line: Standard push 5.5–6.5 h | Minimum defence 3.5 h (must secure: 811 retrieval + M1 + English reading).

Warm-Up Retrieval Checklist (5–10 min): without notes, write on scratch paper:

☐ $\frac{d}{dx} \int_0^{g(x)} f(t) dt = f(g(x))g'(x)$ ☐ Annihilation condition: $\hat{a}|0\rangle = 0 \implies \hat{a}\psi_0(x) = 0$ ☐ Ladder action: $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle, \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$

Warm-Up Retrieval Workspace | 白纸破冰草稿区

5–10 分钟闭卷盲写: 在下方空白区域手推上述 3 个核心公式/边界条件, 快速越过启动能垒。

§2 Core Problem Set

Complete independently under timed conditions

Problem M1 | 604 Calculus: variable-upper limit

Suggested time: 25 min

求极限

$$\lim_{x \rightarrow 0} \frac{\int_0^x (t - \sin t) dt}{x^2(e^{x^2} - 1)},$$

要求写出被积函数的泰勒展开定阶依据及分母等价无穷小替换步骤, 求出最终精确常数数值。

Problem Q1 | 811 Quantum Mechanics: harmonic oscillator wave mechanics

Suggested time: 35 min

质量为 m 的粒子在势场 $V(x) = \frac{1}{2}m\omega^2 x^2$ 中运动。定义无量纲降算符 $\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} + \frac{i}{\sqrt{2m\hbar\omega}}\hat{p}$ 。

- 由基态条件 $\hat{a}\psi_0(x) = 0$, 写出坐标表象下的一阶微分方程, 并求出归一化的基态波函数 $\psi_0(x)$ (已知高斯积分 $\int_{-\infty}^{+\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha}$)。
- 利用坐标算符 $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)$, 计算偶极矩阵元 $\langle m|\hat{x}|n\rangle$, 并指出跃迁选择定则。
- 计算基态下的位置方差 $(\Delta x)^2$ 与动量方差 $(\Delta p)^2$, 并验证不确定性原理 $\Delta x \Delta p = \frac{\hbar}{2}$ 。

Extension | Conceptual Analysis

Optional · Suggested time: 10 min

M1: 为何可将被积函数展开后直接逐项积分? 依据何种定理? Q1: 从宇称对称性说明为何 $\langle n|\hat{x}|n\rangle \equiv 0$ 。
主问题独立作答合计不超过 80 分钟。

§3 Model Solutions & Commentary

Complete your independent attempt on scratch paper before consulting this page.

M1 · Model Solution & Commentary

Strategy: 被积函数包含相消项 $t - \sin t$, 先对其做 Taylor 展开定阶; 分母是乘积形式, 可用等价无穷小快速化简。

Derivation: 当 $t \rightarrow 0$ 时, $\sin t = t - \frac{t^3}{6} + o(t^3) \implies t - \sin t = \frac{t^3}{6} + o(t^3)$ 。

分子定限积分: $\int_0^x (t - \sin t) dt = \int_0^x \left(\frac{t^3}{6} + o(t^3) \right) dt = \frac{x^4}{24} + o(x^4)$ 。

分母乘积项等价代换: $x^2(e^{x^2} - 1) \sim x^2 \cdot x^2 = x^4$ ($x \rightarrow 0$)。故极限为:

$$\lim_{x \rightarrow 0} \frac{\int_0^x (t - \sin t) dt}{x^2(e^{x^2} - 1)} = \lim_{x \rightarrow 0} \frac{\frac{x^4}{24} + o(x^4)}{x^4} = \frac{1}{24}.$$

[Common Error] 被积函数展开到 t^3 阶, 定限积分后直接升至 x^4 阶, 恰与分母 x^4 同阶相消。切忌在被积函数内部盲目只保留到一次项。

Q1 · Model Solution & Commentary

Physical Picture: 降算符作用在基态上为零矢; 坐标算符仅含一次阶梯算符, 跃迁仅在相邻能级发生 ($\Delta n = \pm 1$)。

Derivation: (1) 动量算符 $\hat{p} = -i\hbar \frac{d}{dx}$ 代入 $\hat{a}\psi_0 = 0$: $\left(\sqrt{\frac{m\omega}{2\hbar}}x + \frac{\hbar}{\sqrt{2m\hbar\omega}} \frac{d}{dx} \right) \psi_0 = 0 \implies \frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar} x \psi_0$ 。

分离变量积分得 $\psi_0(x) = A e^{-\frac{m\omega}{2\hbar} x^2}$ 。由归一化 $\int_{-\infty}^{+\infty} |\psi_0|^2 dx = A^2 \sqrt{\frac{\pi\hbar}{m\omega}} = 1 \implies A = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4}$:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar} x^2\right).$$

(2) 利用 $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)$: $\langle m | \hat{x} | n \rangle = \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n}\delta_{m,n-1} + \sqrt{n+1}\delta_{m,n+1})$ 。

跃迁选择定则为: $\Delta n = m - n = \pm 1$ (仅当 $m = n \pm 1$ 时矩阵元非零)。

(3) 基态 $|0\rangle$ 下: $\langle 0 | \hat{x} | 0 \rangle = 0$, $\langle 0 | \hat{p} | 0 \rangle = 0$ 。方差为 $(\Delta x)^2 = \frac{\hbar}{2m\omega}$, $(\Delta p)^2 = \frac{m\hbar\omega}{2}$ 。

不确定度乘积为: $\Delta x \Delta p = \sqrt{\frac{\hbar}{2m\omega}} \cdot \sqrt{\frac{m\hbar\omega}{2}} = \frac{\hbar}{2}$, 精确达到 Heisenberg 理论下限!

[Checkpoint] 归一化系数为 $(m\omega/\pi\hbar)^{1/4}$ (四分之一次幂)。选择定则矩阵元仅两项非零, 基态波包为最小测不准高斯波包。

§4 Daily Debrief & Reflection

Post-Match Review (fill in before sleep, 5 min)

604 Calculus	Time spent: _____ min M1 status: ☐ Independent ☐ Used hints ☐ Incomplete Error type: ☐ Knowledge gap ☐ Modelling ☐ Logic ☐ Calculation ☐ Time
811 QM	Time spent: _____ min Q1 status: ☐ Independent ☐ Used hints ☐ Incomplete Error type: ☐ Knowledge gap ☐ Physical model ☐ Operator algebra ☐ Calculation ☐ Time
Planning	Spaced retrieval in 3 days: 9月4日重做变上限积分定阶与基态波函数一阶微分推导 Today's assessment: ☐ Peak performance ☐ Steady progress ☐ Minimum defence mode

Key Derivation Insights & Traps | 突破口与避坑沉淀

Core Breakthrough (今日核心突破口与关键一步):

Fatal Trap & Common Error (致命陷阱与避坑警示):

Day +3 Action Item (3天后间隔重做提取的具体题号与断点):
